ANOVA (part II)

TEST ABOUT THE EQUALITY OF THE MEANS

Testing equality of the means is equivalent to testing

$\begin{cases} H_0 : \beta_2 = \beta_3 = \dots = \beta_G = 0 \\ H_1 : \overline{H_0} \end{cases}$ test about the overall significance

We used $F = \dfrac{\tilde{\Sigma}^2 - \hat{\Sigma}^2}{\hat{\Sigma}^2} \cdot \dfrac{N-G}{G-1} \overset{H_0}{\sim} F_{G-1, N-G}$

What are $\tilde{\Sigma}^2$ and $\hat{\Sigma}^2$ here?

- $\tilde{\Sigma}^2$ estimator under H_0:

  under H_0 the model we consider is: $\underline{Y} = \beta_1 \cdot \underline{1} + \underline{\varepsilon}$ NULL MODEL

  $\Rightarrow \hat{\beta}_1 = \bar{y}$ overall mean

  Hence, $\tilde{\sigma}^2 = \frac{1}{N} \sum_{g=1}^{G} \sum_{i=1}^{n_g} (y_{ig} - \bar{y})^2 = \frac{SST}{N}$

  and the estimator $\tilde{\Sigma}^2 = \frac{1}{N} \sum_{g=1}^{G} \sum_{i=1}^{n_g} (Y_{ig} - \overline{Y})^2$

- $\hat{\Sigma}^2$ estimator under H_1:

  under H_1 we have the full model (G covariates)

  we estimate $\hat{y}_{ig} = \bar{y}_g$ for $g = 1, \dots, G$ and $i = 1, \dots, n_g$

  Hence, $\hat{\sigma}^2 = \frac{1}{N} \sum_{g=1}^{G} \sum_{i=1}^{n_g} (y_{ig} - \bar{y}_g)^2 = \frac{SSE}{N}$

  and the estimator is $\hat{\Sigma}^2 = \frac{1}{N} \sum_{g=1}^{G} \sum_{i=1}^{n_g} (Y_{ig} - \overline{Y}_g)^2$

Hence the test statistic can be written as:

$$F = \frac{\tilde{\Sigma}^2 - \hat{\Sigma}^2}{\hat{\Sigma}^2} \cdot \frac{N-G}{G-1} =$$

$$= \frac{SST - SSE}{SSE} \cdot \frac{N-G}{G-1} =$$

$$= \frac{SSR}{SSE} \cdot \frac{N-G}{G-1} =$$

$$= \frac{R^2}{1-R^2} \cdot \frac{N-G}{G-1} \overset{H_0}{\sim} F_{G-1, N-G}$$

SUM OF SQUARES DECOMPOSITION

consider G groups, and n_g observations in each group:

$Y_{ig} \sim N(\mu_g, \sigma^2)$ independent for $i = 1, \dots, n_g$ and $g = 1, \dots, G$

Let $N = \sum_{g=1}^{G} n_g$ total sample size

- The group-specific means are

$$\bar{y}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} y_{ig} \quad g = 1, \dots, G$$

- The overall mean is

$$\bar{y} = \frac{1}{N} \sum_{g=1}^{G} \sum_{i=1}^{n_g} y_{ig} = \frac{1}{N} \sum_{g=1}^{G} n_g \bar{y}_g$$

- The group-specific estimates of the variance are

$$s_g^2 = \frac{1}{n_g - 1} \sum_{i=1}^{n_g} (y_{ig} - \bar{y}_g)^2 \qquad g = 1, \dots, G$$

The partition of the sum of squares in the linear model was $\overset{SST}{\sum_{i=1}^{n} (y_i - \bar{y})^2} = \overset{SSR}{\sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2} + \overset{SSE}{\sum_{i=1}^{n} (y_i - \hat{y}_i)}$

We can specify it for this setting:

The total sum of squares here is $SST = \sum_{g=1}^{G} \sum_{i=1}^{n_g} (y_{ig} - \bar{y})^2$

$$\sum_{g=1}^{G} \sum_{i=1}^{n_g} (y_{ig} - \bar{y})^2 = \sum_{g=1}^{G} \sum_{i=1}^{n_g} (y_{ig} - \bar{y}_g + \bar{y}_g - \bar{y})^2$$

$$= \sum_{g=1}^{G} \sum_{i=1}^{n_g} \left[(y_{ig} - \bar{y}_g)^2 + (\bar{y}_g - \bar{y})^2 + 2(y_{ig} - \bar{y}_g)(\bar{y}_g - \bar{y}) \right]$$

$$= \sum_{g=1}^{G} \sum_{i=1}^{n_g} (y_{ig} - \bar{y}_g)^2 + \sum_{g=1}^{G} \sum_{i=1}^{n_g} (\bar{y}_g - \bar{y})^2 + 2 \sum_{g=1}^{G} \sum_{i=1}^{n_g} (\bar{y}_g - \bar{y})(y_{ig} - \bar{y}_g)$$

$$= \underbrace{\sum_{g=1}^{G} \sum_{i=1}^{n_g} (y_{ig} - \bar{y}_g)^2}_{= (n_g - 1) s_g^2} + \sum_{g=1}^{G} n_g (\bar{y}_g - \bar{y})^2 + 2 \sum_{g=1}^{G} (\bar{y}_g - \bar{y}) \underbrace{\sum_{i=1}^{n_g} (y_{ig} - \bar{y}_g)}_{= 0}$$

$$= \sum_{g=1}^{G} (n_g - 1) s_g^2 + \sum_{g=1}^{G} n_g (\bar{y}_g - \bar{y})^2$$

Hence we get

$$\underbrace{\sum_{g=1}^{G} \sum_{i=1}^{n_g} (y_{ig} - \bar{y})^2}_{\text{TOTAL SUM OF SQUARES}} = \underbrace{\sum_{g=1}^{G} (n_g - 1) s_g^2}_{\text{WITHIN-GROUP VARIABILITY}} + \underbrace{\sum_{g=1}^{G} n_g (\bar{y}_g - \bar{y})^2}_{\text{BETWEEN-GROUP VARIABILITY}}$$

↝ "ANOVA": Analysis of variance

Total sum of squares: deviations of each observation from the overall mean

Within-group sum of squares: deviations of each observation from the corresponding group-specific mean

Between-group sum of squares: deviations of each group-specific mean from the overall mean

Moreover, we have seen that $\bar{y}_g = \hat{y}_{ig}$ for $i = 1, \dots, n_g$

that is, the predicted values are the group-specific means

Thus

$$\sum_{g=1}^{G} (n_g - 1) s_g^2 = \sum_{g=1}^{G} \sum_{i=1}^{n_g} (y_{ig} - \bar{y}_g)^2 = \sum_{g=1}^{G} \sum_{i=1}^{n_g} (y_{ig} - \hat{y}_{ig})^2 \quad \text{ERROR SUM OF SQUARES}$$

$$\sum_{g=1}^{G} n_g (\bar{y}_g - \bar{y})^2 = \sum_{g=1}^{G} \sum_{i=1}^{n_g} (\bar{y}_g - \bar{y})^2 = \sum_{g=1}^{G} \sum_{i=1}^{n_g} (\hat{y}_{ig} - \bar{y})^2 \quad \text{REGRESSION SUM OF SQUARES}$$

Hence, the F test can also be expressed as

$$F = \frac{\text{BETWEEN-GROUP S.S.}}{\text{WITHIN-GROUP S.S.}} \cdot \frac{N-G}{G-1}$$